

39. Let the ten's digit be  $x$ . Then, unit's digit =  $x + 3$ .

$$\text{Number} = 10x + (x + 3) = 11x + 3.$$

$$\text{Sum of digits} = x + (x + 3) = 2x + 3.$$

$$\therefore \frac{11x+3}{2x+3} = \frac{4}{1} \Leftrightarrow 11x + 3 = 8x + 12$$

$$\Leftrightarrow 3x = 9 \Leftrightarrow x = 3.$$

$$\text{Hence, required number} = 11x + 3 = 11 \times 3 + 3 = 36.$$

40. Let the numbers be  $3x$  and  $4x$ .

$$\text{Then, } (4x)^2 = 8 \times (3x)^2 - 224$$

$$\Leftrightarrow 16x^2 = 72x^2 - 224$$

$$\Leftrightarrow 56x^2 = 224 \Leftrightarrow x^2 = 4$$

$$\Leftrightarrow x = 2.$$

So, the numbers are 6 and 8.

41. Let the numbers be  $4x$  and  $7x$ .

$$\text{Then, } \frac{4x+4}{7x+4} = \frac{3}{5} \Leftrightarrow 5(4x+4) = 3(7x+4) \Leftrightarrow x = 8.$$

$$\therefore \text{Larger number} = 7x = 56.$$

42. Let the second number be  $x$ . Then, first number =  $2x$

$$\text{and third number} = \frac{2x}{3}.$$

$$\therefore 2x + x + \frac{2x}{3} = 264 \Leftrightarrow \frac{11x}{3} = 264$$

$$\Leftrightarrow x = \left( \frac{264 \times 3}{11} \right) = 72.$$

43. Let the numbers be  $x$  and  $(22 - x)$ . Then,  $5x = 6(22 - x)$

$$\Leftrightarrow 11x = 132 \Leftrightarrow x = 12.$$

So, the numbers are 12 and 10.

44. Let the numbers be  $x$  and  $y$ . Then,  $\frac{1}{5}x = \frac{5}{8}y \Leftrightarrow y = \frac{8}{25}x$ .

$$\text{Now, } x + 35 = 4y \Leftrightarrow x + 35 = \frac{32}{25}x$$

$$\Leftrightarrow \frac{7}{25}x = 35 \Leftrightarrow x = \left( \frac{35 \times 25}{7} \right) = 125.$$

$$\therefore \text{Second number} = y = \frac{8}{25}x = \left( \frac{8}{25} \times 125 \right) = 40.$$

45. Let the numbers be  $x$  and  $y$ .

$$\text{Then, } x + y = 25 \text{ and } x - y = 13.$$

$$4xy = (x + y)^2 - (x - y)^2 = (25)^2 - (13)^2$$

$$= 625 - 169 = 456 \Rightarrow xy = 114.$$

46. Let the numbers be  $x$  and  $y$ . Then,

$$x + y = 33 \quad \dots(i)$$

$$\text{and } x - y = 15 \quad \dots(ii)$$

Solving (i) and (ii), we get :  $x = 24, y = 9$ .

$$\therefore \text{Smaller number} = 9.$$

47. Let the numbers be  $x$  and  $y$ .

$$\text{Then, } \frac{x+y}{x-y} = \frac{40}{4} = 10 \Leftrightarrow (x+y) = 10(x-y)$$

$$\Leftrightarrow 9x = 11y \Leftrightarrow \frac{x}{y} = \frac{11}{9}.$$

48. Let the numbers be  $x$  and  $(28 - x)$ .

$$\text{Then, } x(28 - x) = 192 \Leftrightarrow x^2 - 28x + 192 = 0$$

$$\Leftrightarrow (x - 16)(x - 12) = 0$$

$$\Leftrightarrow x = 16 \text{ or } x = 12.$$

So, the numbers are 16 and 12.

49. Let the numbers be  $x$  and  $y$ . Then,

$$2x + 3y = 100 \quad \dots(i)$$

$$\text{and } 3x + 2y = 120 \quad \dots(ii)$$

Adding (i) and (ii), we get :  $5x + 5y = 220$

$$\text{or } x + y = 44 \quad \dots(iii)$$

Subtracting (i) from (ii); we get :  $x - y = 20 \quad \dots(iv)$

Adding (iii) and (iv), we get:  $2x = 64$  or  $x = 32$ .

Putting  $x = 32$  in (iii), we get :  $y = 12$ .

Hence, larger number = 32.

50. Let the numbers be  $x$  and  $y$ . Then,

$$xy = 1092 \quad \dots(i)$$

$$\text{And, } (x + y) - (x - y) = 42$$

$$\Leftrightarrow 2y = 42$$

$$\Leftrightarrow y = 21.$$

$$\text{Putting } y = 21 \text{ in (i), we get : } x = \frac{1092}{21} = 52.$$

Hence, greater number = 52.

51. Let the integers be  $x$  and  $(x + 5)$ . Then,

$$x(x + 5) = 500 \Leftrightarrow x^2 + 5x - 500 = 0$$

$$\Leftrightarrow (x + 25)(x - 20) = 0$$

$$\Leftrightarrow x = 20.$$

So, the numbers are 20 and 25.

52. Let the numbers be  $x$  and  $y$ .

$$\text{Then, } x - y = 5 \text{ and } xy = 336.$$

$$(x + y)^2 = (x - y)^2 + 4xy = 25 + 4 \times 336 = 1369$$

$$\Rightarrow x + y = \sqrt{1369} = 37.$$

53. Since  $1 < 1 + x$ , so one of the numbers is 1.

54. Let the numbers be  $x$  and  $y$ .

$$\text{Then, } xy = 9375 \text{ and } \frac{x}{y} = 15.$$

$$\frac{xy}{(x/y)} = \frac{9375}{15} \Leftrightarrow y^2 = 625 \Leftrightarrow y = 25$$

$$\Rightarrow x = 15y = (15 \times 25) = 375.$$

$$\therefore \text{Sum of the numbers} = 375 + 25 = 400.$$

55. Let the numbers be  $x$  and  $(x + 1365)$ .

$$\text{Then, } x + 1365 = 6x + 15 \Leftrightarrow 5x = 1350 \Leftrightarrow x = 270.$$

56. Let the numbers be  $x$  and  $(x + 16)$ .

$$\text{Then, } \frac{x}{3} - \frac{(x+16)}{7} = 4 \Leftrightarrow 7x - 3(x+16)$$

$$= 84 \Leftrightarrow 4x = 84 + 48$$

$$= 132 \Leftrightarrow x = 33.$$

Hence, the numbers are 33 and 49.

57. Let the numbers be  $x$  and  $y$ .

$$\text{Then, } x + y = 40 \text{ and } xy = 375.$$

$$\therefore \frac{1}{x} + \frac{1}{y} = \frac{x+y}{xy} = \frac{40}{375} = \frac{8}{75}.$$